



GAUTENG PROVINCE
EDUCATION
REPUBLIC OF SOUTH AFRICA

PROVINCIAL EXAMINATION

NOVEMBER 2023

GRADE 6

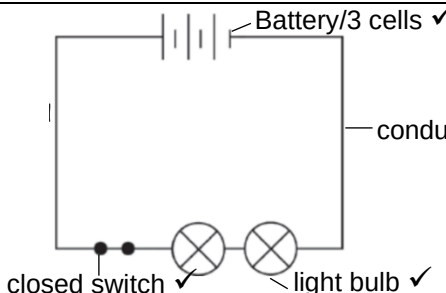
MARKING GUIDELINES

NATURAL SCIENCE AND TECHNOLOGY

6 pages

SECTION A	
QUESTION 1	MARKS
1.1 Three (3) ✓	1
1.2 To provide power to the circuit ✓	1
1.3 Yes ✓	1
1.4 Light energy ✓	1
1.5 Light bulb ✓	1
	[5]
QUESTION 2	
2.1 C ✓	1
2.2 D ✓	1
2.3 C ✓	1
2.4 A ✓	1
2.5 B ✓	1
	[5]
QUESTION 3	
3.1 B ✓	1
3.2 C ✓	1
3.3 A ✓	1
3.4 E ✓	1
3.5 D ✓	1
	[5]
QUESTION 4	
4.1 True ✓	1
4.2 True ✓	1
4.3 True ✓	1
4.4 True ✓	1
4.5 False ✓	1
	[5]

QUESTION 5	
5.1 Revolution ✓	1
5.2 Moon ✓	1
5.3 Axis ✓	1
5.4 Crater ✓	1
5.5 Telescope ✓	1
	[5]
QUESTION 6	
6.1 Tungsten Filament ✓	1
6.2 Filament ✓	1
6.3 Cap ✓	1
6.4 Contact wire ✓	1
6.5 Electrical contact ✓	1
	[5]
TOTAL SECTION A:	
	30

SECTION B		
QUESTION 7		MARKS
C ✓		5
B ✓		
D ✓		
E ✓		
A ✓		
		[5]
QUESTION 8		
8.1 (a) Electric stoves and geysers are high-power appliances that require more energy to generate heat. ✓ Appliances that are used for heating use more electrical energy and will therefore cost more. ✓		2
(b) 1. Use a gas stove. ✓ 2. Use a solar geyser. ✓		2
(c) Solar energy ✓		1
8.2 (a) The electrical wire is exposed. ✓ (b) Cover the exposed part with insulation tape. ✓		2
		[7]
QUESTION 9		
9.1	Marking Criteria • Correct symbols must be used. • Each component must be labelled. • Wires to be drawn using a ruler. • Switch must be closed. ✓✓✓✓ • $\frac{1}{2}$ mark per label and 2 marks for complete/correct diagram	
		4
9.2	INPUT Electrical energy from the battery ✓ ➡ PROCESS Electrical charges move through the circuit. ➡ OUTPUT Electrical energy changes to light energy. ✓	
		2
9.3	Circuit B will produce an output. ✓ There are no gaps in the circuit./The switch is closed. ✓✓	3
		[9]
TOTAL SECTION B:		21

SECTION C	
QUESTION 10	MARKS
10.1 Take pictures of the surface./Study the rocks and soil./Collect rock samples./ Transport astronauts./Save their energy and oxygen supply. ✓✓ (Accept any suitable answer)	2
10.2 It is an unsafe environment./Mars is too far (distance). ✓	1
	[3]
QUESTION 11	
11.1 It should be away from people ✓ and from broadcasting, FM and cellphone towers to prevent interference. ✓	2
11.2 It will be 50 times more sensitive ✓ and be able to survey the sky 10 000 times faster ✓ than other telescopes.	2
11.3 The bottom of the glass bottle will represent/be used as a magnifying glass for the telescope. ✓✓	2
	[6]
TOTAL SECTION C:	
	9
TOTAL:	
	60

**NATURAL SCIENCE AND TECHNOLOGY
ASSESSMENT FRAMEWORK
GRADE 6
PROVINCIAL EXAM – NOVEMBER 2023
DURATION: 1½ hours**

SECTION A	MARKS
QUESTION 1: ANSWER QUESTIONS	5
QUESTION 2: MULTIPLE-CHOICE QUESTIONS	5
QUESTION 3: MATCHING ITEMS	5
QUESTION 4: TRUE OR FALSE	5
QUESTION 5: ONE WORD (SCIENTIFIC TERM)	5
QUESTION 6: LABELLING	5
TOTAL SECTION A:	30
SECTION B	MARKS
QUESTION 7: REARRANGE IN CORRECT ORDER	5
QUESTION 8: USE OF KNOWLEDGE	7
QUESTION 9: USE OF KNOWLEDGE, ILLUSTRATE, COMPARE	9
TOTAL SECTION B:	21
SECTION C	MARKS
QUESTION 10: CASE STUDY/INVESTIGATION	6
QUESTION 11: REASONING	3
TOTAL SECTION C:	9
TOTAL:	60